A Seamless Pedestrian Localization System Based on GNSS/IMU/UWB/MAP Integration

Shuning Zhang, Hongjing Tang, Liangming Chen, *Senior Member, IEEE*, and Yufeng Gao

***Abstract*—Pedestrian localization has broad applications. However, achieving seamless positioning in complex environments remains a significant challenge due to factors like environmental occlusion and indoor-outdoor transitions. This article proposes a combined localization framework using global navigation satellite system (GNSS), inertial measurement unit (IMU), ultrawideband (UWB), and MAP data to address this issue. In the framework, a three-step GNSS localization method is proposed to enhance accuracy and stability by eliminating nonline-of-sight (NLOS) effects and filtering anomalies. Inertial navigation system (INS) and pedestrian dead reckoning (PDR) models are integrated to improve local precision and stability. UWB, known for its superior localization capabilities, has been integrated into some smartphones and is promising for large-scale indoor applications. This article presents a gradient descent-based optimal anchor deployment algorithm and localization methods for varying anchor quantities. Additionally, a nonstationary Markov model fuses map data for trajectory prediction and correction. Finally, anomaly detection filtering and low-pass filtering are applied to the corrected trajectories, yielding high accuracy and stability. Experiments on the Honor V20 and iQOO U3 smartphones showed significant improvements compared to the baseline algorithm. Outdoor root mean square error (RMSE) was reduced by 44.4% (10.83 m) for the Honor V20 and 40.0% (10.77 m) for the iQOO U3. Additionally, outdoor standard deviation (STD) improved by 43.9% (4.80 m) and 43.5% (5.50 m), respectively. In transitional areas, RMSE improved by 4.61% (10.77 m) and 19.6% (4.76 m), respectively. Indoor RMSE also showed improvements, with the Honor V20 reducing it by 8.00% (3.79 m) and the iQOO U3 by 5.78% (3.91 m).**

***Index Terms*—Global navigation satellite system (GNSS), inertial navigation system (INS), map constraints and path prediction, Markov model, pedestrian dead reckoning (PDR), ultrawideband (UWB).**

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I. Introduction

NAVIGATION and positioning applications are integral to modern life, influencing industries, such as agriculture, healthcare, and services. Satellite navigation systems dominate outdoor positioning due to their reliability, global coverage, real-time capabilities, and high accuracy [1], [2]. Indoor positioning technologies include Wi-Fi [3], [4], [5], RFID [6], Bluetooth [7], [8], ultrawideband (UWB) [9], [10], [11], visual systems [12], [13], visible light [14], and 5G [15], [16]. Despite their variety, many face significant limitations that hinder large-scale applications. UWB stands out for its precision, low power consumption, and real-time capabilities [17], emerging as a potential mainstream solution. Phone manufacturers like Apple and Samsung have already integrated UWB chips into their products [18], paving the way for large-scale indoor deployments in environments, such as malls and hospitals. One of UWB's key applications is seamless indoor-outdoor pedestrian localization, which is crucial for daily tasks like commuting and indoor navigation [19].

In summary, current solutions integrate inertial measurement unit (IMU), global navigation satellite system (GNSS), Wi-Fi, and Bluetooth sensors, utilizing algorithms, such as pedestrian dead reckoning (PDR), Kalman filtering, and machine learning. IMU sensors are valued for their short-term accuracy and resistance to interference [20], [21], though they suffer from error accumulation. Therefore, it is usually corrected in conjunction with GNSS [22], [23]. However, the accuracy of GNSS on smartphones typically ranges from several meters to tens of meters, with performance significantly degrading in semi-obstructed environments. For indoor scenarios, localization methods relies on Wi-Fi or IMU/Wi-Fi fusion, with alternatives like Bluetooth and visible light [24], [25], [26], [27], [28]. However, these technologies often encounter large errors, incomplete coverage, or complex infrastructure requirements. UWB, with its submeter accuracy and broad coverage, offers a promising solution [29].

To enhance pedestrian localization, researchers have integrated PDR models with techniques like zero velocity update (ZUPT) [30], [31] and step estimation [32], [33], using Kalman filtering to correct errors. Machine learning approaches, such as RIDI [34] and IONet [35], also show potential but require large training datasets, limiting their generalizability. For GNSS-based systems, Kalman filtering combined with inertial navigation system (INS) and PDR models enhances accuracy [22], [23], while map information aids in adjusting results for better reliability [30], [36], [37].

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Moreover, advanced GNSS techniques like DGNSS [38], PPP [39], and RTK [40] further improve accuracy.

The seamless integration of indoor and outdoor positioning remains a key challenge for pedestrian localization. Current mainstream strategies are as follows.

1) *Multisensor Fusion [36], [41], [42], [43], [44], [45]:* Different positioning systems can be used in various regions to achieve comprehensive coverage for seamless localization.
2) *Environment-Based Localization System Switching [42], [45], [46]:* This method relies on changes in sensor signals to adopt the most suitable positioning model for each environment, facilitating seamless localization.
3) *Enhanced Coverage in Transition Zones [43]:* Additional positioning systems, such as UWB, can be integrated into transition areas to improve performance.
4) *Map Model-Assisted Correction [4], [30], [37]:* Utilizing electronic maps for positioning adjustments can effectively suppress localization drift and enhance continuity, thereby achieving seamless localization.

Among these, multisensor fusion navigation is the most common approach, typically combining outdoor systems (GNSS, maps), indoor systems (WIFI, Bluetooth, and UWB), and transitional systems (IMU, VIO, and landmarks). Popular combinations include GNSS/WIFI/INS [36], [41], GNSS/Bluetooth/INS [42], PPP/UWB/INS [43], and RTK/UWB/VIO [44]. Additional sensors like light and geomagnetic sensors [45], and LiDAR are also used. Integration across systems is mainly achieved through EKF filtering, combining loosely or tightly coupled (TC) methods. It also involves UKF [47], FKF [48], and factor graphs [22], [39], [49].

The main goal of environment detection switching is to improve localization in specific scenarios. Light sensors can detect changes in light intensity to identify indoor-outdoor environment but require robust switching mechanisms to avoid interference from windows or corridors [45]. Signal strength fluctuations in WIFI and GNSS can also be used for environmental switching [42], [46], but intelligent algorithms are needed to handle similar characteristics caused by trees or tall buildings. In addition, hidden Markov models (HMM) [42], [46] are commonly employed for scene transition detection.

Enhanced transition zone coverage and map-assisted corrections are usually integrated with multisensor systems. Relying solely on transition area coverage cannot provide complete indoor-outdoor positioning, so it needs to be combined with other indoor or outdoor positioning systems. For example, in [43], UWB is used to enhance transition area coverage and improve positioning performance. Moreover, the map model itself cannot perform positioning but primarily serves to correct and constrain the system [4], [30], [37].

Currently, multisource systems enable high-precision seamless localization for vehicles [43], [44]. However, pedestrian seamless localization still faces accuracy limitations, especially indoors, where high-precision methods are lacking. Pedestrian systems also struggle with robustness during indoor-outdoor transitions, causing sharp accuracy drops. This article aims to develop a high-precision, robust, and seamless pedestrian localization system and its associated algorithms. Our main contributions are as follows.

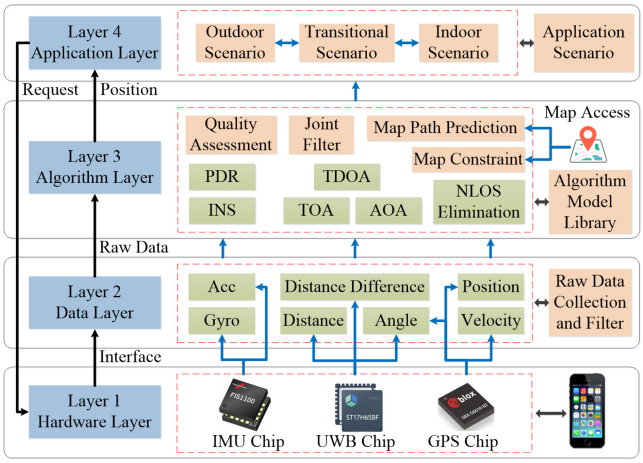


Fig. 1. Overall application architecture of the system.

1) A high-precision seamless pedestrian localization system based on GNSS/IMU/UWB/MAP was structured. It featured an integrated algorithm flow that combined factor graphs, nonline-of-sight (NLOS) recognition, INS, PDR, map constraints, path prediction, and joint filtering.
2) A novel 3D-NLOS Discrimination Line (3D-NLOS-DL) was developed for efficient identification of NLOS signals, coupled with time-series anomaly filtering to enhance the accuracy and robustness of single GNSS.
3) The spatially weighted PDOP (SWPDOP) concept was developed for optimal UWB anchor deployment, utilizing spatial weighting and gradient descent for effective anchor placement.
4) A map constraint formula and a path prediction algorithm based on a conditional extended Markov model were developed, significantly improving accuracy.

The remainder of this article is structured as follows. Section II covers the system model, Section III details the algorithm, Section IV presents experimental results and comparisons, and Section V concludes with future research directions.

II. Overview of the System

Based on the analysis in Section I, this study proposes a seamless pedestrian localization model combining GNSS, UWB, and IMU sensors on a mobile application platform, as shown in Fig. 1.

The system architecture has four layers. The first is the hardware layer, which includes sensor chips like the IMU, UWB, and GNSS chips on the smartphone, connected to the data layer via hardware interfaces. The second is the data layer, where raw data from sensors is collected and prefiltered. The IMU provides tri-axial acceleration and angular velocity data. UWB provides distance and angle data. GNSS offers pseudorange and satellite count data, which are processed into navigation data. Additionally, prefiltering algorithms like the Kalman filter removes outliers and reduces NLOS effects.

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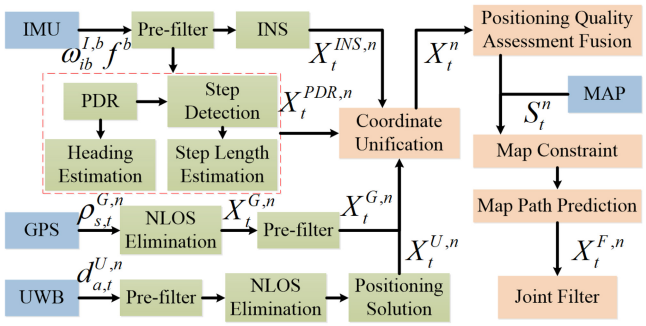


Fig. 2. Data processing algorithm flowchart.

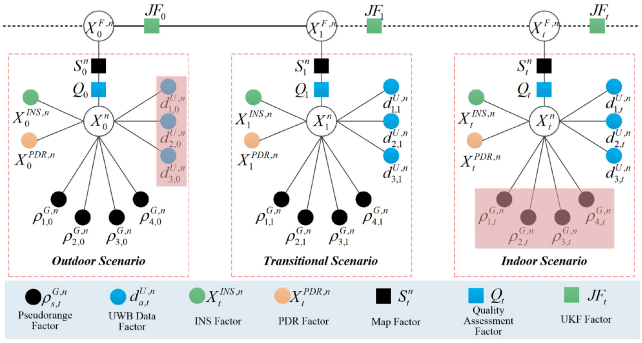


Fig. 3. GNSS/UWB/IMU/MAP factor graph optimization model.

The third layer is the algorithm layer, featuring localization algorithms. For IMU data, algorithms like PDR and INS are used, while UWB data employs TOA and TDOA methods. GNSS data involves advanced algorithms like NLOS recognition and RTK. Map-assisted algorithms for constraint and prediction are also included. Fusion algorithms, such as quality assessment fusion and joint filter are used at this layer, where combining the right algorithms is key to achieving high accuracy and robustness.

The fourth layer is the application layer, supporting scenarios like outdoor, transition zones, and indoor environments. The process begins with the application layer requesting data, followed by data collection and prefiltering in the data layer, algorithm integration in the algorithm layer, and finally, seamless localization results are displayed.

Fig. 2 depicts the proposed localization algorithm flow. Data sources (IMU, GNSS, UWB, and MAP) are shown in blue, basic algorithms in green, and upper-layer fusion algorithms in orange. IMU data undergoes INS and PDR processing to produce state variables X^{INS} and X^{PDR}, GNSS data are processed to yield X^G, and UWB data generates X^U. All these state variables are unified into a state matrix X^n, followed by quality assessment-based weight fusion. The 3-D map data S^n is used for localization constraints and path prediction, employing a nonstationary Markov model. Joint Filters are then applied to improve accuracy and stability.

III. Factor Graph Optimization-Based Seamless Pedestrian Localization Algorithm

Factor graph optimization was initially developed to express probabilistic relationships among different variable nodes, but its applications have expanded beyond probability, encompassing functions like Kalman filtering, INS, and PDR [22], [49]. This flexibility allows factor graphs to clearly express the iterative relationships of variables in integrated navigation systems. While Fig. 2 illustrates the data processing flow for a single moment, factor graphs extend the system's capability by incorporating historical data for optimization purposes. To better visualize the optimization process, a factor graph model is employed, as shown in Fig. 3.

Fig. 3 depicts the factor graph optimization model based on the GNSS/UWB/IMU/MAP architecture. In this model, the hollow circles represent state nodes, with each time step having two state nodes. X_t denotes the concatenated state matrix computed from the four systems (GNSS, UWB, IMU, and MAP), while X_t^F represents the fused state matrix at the current time.

The black circles indicate pseudorange factors, which are the signals received from different satellites by the smartphone at various time steps. Blue circles correspond to measurement data factors received from UWB tags, recorded by anchors at different moments. Green circles represent INS function factors, while orange circles correspond to PDR function factors. The black rectangles represent 3-D map model factors, which can be divided into map constraint factors and path prediction factors. The blue rectangle denotes the quality assessment factor, which converts the concatenated state matrix into the fused state matrix.

The fused state matrices over time form a time series, which is then smoothed and filtered using Joint Filter factors. Red shading in the figure highlights factors that may be unavailable depending on the environment. For instance, in outdoor settings where no UWB anchors are present, UWB factors are absent. In indoor environments, GNSS pseudorange factors are not accessible. However, in transition zones, all factors are available, improving scene recognition and enabling seamless positioning.

A. INS Model Factor

INS navigation, which does not rely on external information for navigation, has received extensive attention and research since its inception. It has now become a fundamental navigation model that is integrated with various emerging systems. Therefore, the differential equations in the n-frame are directly presented as follows in (1) [50]

$$\dot{\boldsymbol{C}}_b^n = \boldsymbol{C}_b^n\left(\boldsymbol{\omega}_{nb}^b\times\right) = \boldsymbol{C}_b^n\left(\boldsymbol{\omega}_{ib}^b\times\right) - \left(\boldsymbol{\omega}_{\mathrm{in}}^n\times\right)\boldsymbol{C}_b^n \tag{1a}$$

$$\dot{\boldsymbol{v}}^n = \boldsymbol{C}_b^n\boldsymbol{f}^b - \left(2\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n\right)\times\boldsymbol{v}^n + \boldsymbol{g}_p^n \tag{1b}$$

$$\dot{\mathbf{P}}_t^n = \boldsymbol{M}_{pv}\boldsymbol{v}^n = \begin{bmatrix}\dot{\phi} & \dot{\lambda} & \dot{h}\end{bmatrix}^T \tag{1c}$$

$$\boldsymbol{M}_{pv} = \begin{bmatrix} 0 & 1/(R_M+h) & 0 \\ \sec\varphi/(R_N+h) & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{1d}$$

where $\boldsymbol{C}_b^n$, $\boldsymbol{v}^n$, and $\mathbf{P}_t^n$ represent the direction cosine, velocity, and position matrix of the pedestrian in the n-frame, respectively. $\boldsymbol{\omega}_{ib}^b$ and $\boldsymbol{\omega}_{\mathrm{in}}^n$ denote the angular velocities of the b-frame and n-frame relative to the i-frame. $\boldsymbol{f}^b$ represents the

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specific force in the b-frame, and $\boldsymbol{g}_p^n$ denotes the gravitational acceleration in the n-frame. ϕ, λ, and h are the latitude, longitude, and altitude, respectively. $\boldsymbol{M}_{pv}$ is the coefficient matrix, R_M and R_N represent the radii of the meridian and prime vertical, respectively.

B. PDR Model Factor

The PDR model is similar to the INS model and has attracted considerable attention and research from many scholars due to its wide applicability. Its basic principle involves pedestrian's movement and position estimation through motion modeling. Key issues include gait detection, step length estimation, and heading detection. Alcázar-Fernández et al. [28] described a comprehensive 2-D PDR model. To achieve more accurate 3-D seamless positioning, we improve a PDR model based on integrated position feedback correction, with its computational formula provided in (2) [28]

$$\boldsymbol{X_t} = \begin{bmatrix} L_t & \theta_t & \beta_t \end{bmatrix} \tag{2a}$$

$$\boldsymbol{L}_t^n = \begin{bmatrix} L_t \sin\theta_t \cos\beta_t & L_t \sin\theta_t \sin\beta_t & L_t \cos\theta_t \end{bmatrix}^T \tag{2b}$$

$$\mathbf{P}_t^n = \mathbf{P}_{t-1}^n + \boldsymbol{L}_t^n \tag{2c}$$

where $\boldsymbol{X_t}$ represents the PDR parameters calculated from position fusion at time t, including step length L_t, pitch angle θ_t, and yaw angle β_t. $\boldsymbol{L}_t^n$ represents the motion vector in the n-frame, and $\mathbf{P}_t^n$ represents the position at time t in the n-frame.

The specific model parameters for the feedback correction are shown in (3). For pedestrian motion, the step length remains constant over short periods, so it is estimated using historical velocity data. As for the motion angle, INS provides relatively accurate angle estimation, so the current direction is fused with the INS angle using weighted integration to improve the accuracy of the angle estimation

$$\Delta\mathbf{P}_t = \mathbf{P}_t^n - \mathbf{P}_{t-1}^n \tag{3a}$$

$$L = a_L \|\Delta\mathbf{P}_t\| + (1 - a_L) \sum_{i=t-k-1}^{t-1} \frac{\|\Delta\mathbf{P}_i\|}{k} \tag{3b}$$

$$\theta_k = a_\theta \theta_k^{INS} + (1 - a_\theta) \arccos \Delta\mathbf{P}_t(3)/\|\Delta\mathbf{P}_t\| \tag{3c}$$

$$\beta_k = a_\beta \beta_k^{INS} + \left(1 - a_\beta\right) \arccos \Delta\mathbf{P}_t(1,2)/\|\Delta\mathbf{P}_t \tag{3d}$$

where $\mathbf{P}_t^n$ represents the position at time t in the n-frame, $\Delta\mathbf{P}_t$ represents the positional difference vector between two consecutive steps, a_L, a_θ, and a_β denote fusion ratios, all belonging to the interval [0,1], and θ_k^{INS}, β_k^{INS} represent the pitch and yaw angles calculated by INS, respectively.

C. GNSS NLOS Elimination and Positioning Filter Factor

GNSS includes various positioning methods, with pseudorange positioning being the most fundamental and widely used. Pseudorange is the measured distance between the target and the satellite. It is affected by errors, such as satellite orbit and clock biases, ionospheric and tropospheric delays, and multipath effects. While precise ephemeris and clock bias files, along with ionospheric and tropospheric models, address most errors, multipath effects caused by NLOS obstructions lack a unified solution due to their environment-specific nature.

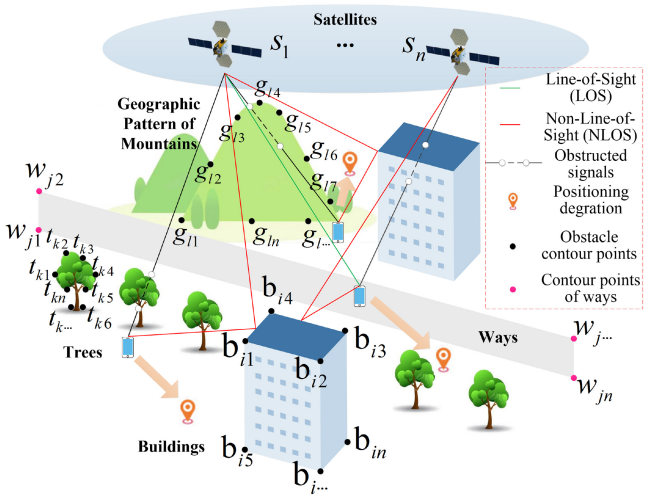


Fig. 4. 3-D map model and extended NLOS signal obstruction model [30].

NLOS-induced pseudorange errors significantly degrade positioning accuracy, making effective NLOS mitigation models essential.

Wang et al. [30] studied the 3-D Mapping-Aided NLOS Signal Detection Algorithm, which is commendable but only addresses building-type obstructions. In practical applications, factors like trees, overpasses, and terrain also frequently impact positioning and should be considered. Based on this, we propose an extended NLOS discrimination model and introduce the concept of the 3D-NLOS Discrimination Line (3D-NLOS-DL), which utilizes 3-D mapping to quickly assess and eliminate NLOS signals in conjunction with satellite elevation angles. Additionally, considering pedestrian motion constraints, we apply time series change detection to filter positioning calculation data, culminating in the proposed 3DMAP-assisted three-step positioning method.

1) 3-D Map Model and Extended NLOS Signal Obstruction Model: The schematic diagram is shown in Fig. 4 [30]. The obstruction sources in Fig. 4 include buildings, trees, terrain, and common geographic elements like paths. This geographic information is encapsulated in the 3-D map structure data $\mathbf{S}_{p,r}^n$, represented by

$$\mathbf{S}_{p,r}^n = \left\{ \cdots \boldsymbol{B}_i^n, \ldots, \boldsymbol{W}_j^n, \ldots, \boldsymbol{T}_k^n, \ldots, \boldsymbol{G}_l^n, \ldots \right\}$$
$$\{i, j, k, l\} \in \mathbb{Z}^+ \wedge \left\{ i \le n_i, j \le n_j, k \le n_k, l \le n_l \right\} \tag{4}$$

where $\mathbf{S}_{p,r}^n$ represents the collection of map structural data contained within a spherical space centered at point p with radius r in the n-frame; $\boldsymbol{B}_i^n$ denotes the 3-D model collection of the ith building, consisting of n_i buildings; $\boldsymbol{W}_j^n$ indicates the 3-D model collection of the jth road, comprising n_j roads; $\boldsymbol{T}_k^n$ signifies the 3-D model collection of the kth tree, containing n_k trees; and $\boldsymbol{G}_l^n$ represents the 3-D model collection of the lth terrain, consisting of n_l terrains.

2) NLOS Discrimination Model: The NLOS discrimination model in [30] uses the elevation and azimuth angles of satellites and building profiles for NLOS determination. While this method is correct, its representation and efficiency can be further improved. We propose that, in an outdoor 3-D

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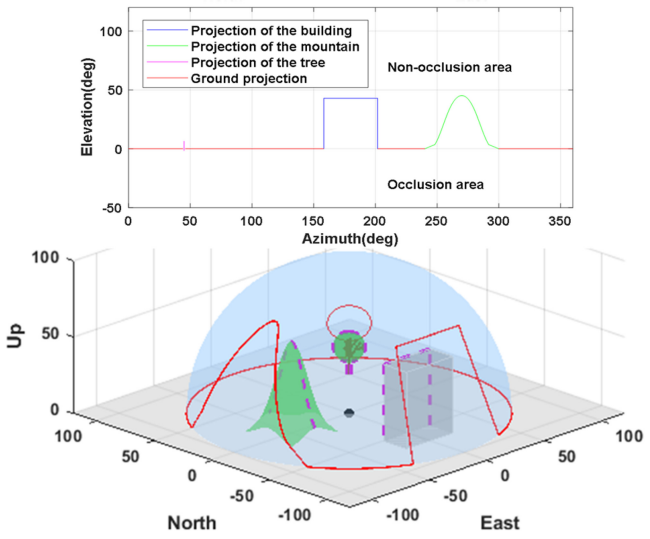


Fig. 5. Diagram of 3D-NLOS-DL principle.

map, any position can be taken as the center of a sphere. Projecting continuously in all directions (360 degrees) creates a continuous curve that divides the sphere into obstructed and unobstructed areas. This curve is defined as the 3D-NLOS-DL, with its principle and mathematical expression shown in Fig. 5.

As shown in Fig. 5, for any point on the map, the 3D-NLOS-DL can be calculated using the projection method, which divides the spatial domain into obstructed and unobstructed areas. To assess any satellite signal for NLOS, it suffices to determine whether it falls within the obstructed or unobstructed area, with the judgment formula provided in

$$E_p^n = f\left(A_p^n\right), A_p^n \in [0, 2\pi], E_p^n \in [0, 2\pi] \tag{5a}$$

$$l_s = \begin{cases} 1|_{\text{LOS}}, & E_s^n > \max\left(f\left(A_s^n\right)\right) \wedge \\ E_s^n < 2\pi - \max\left(f\left(A_s^n + \pi\right)\right) & \\ 0|_{\text{NLOS}}, & E_s^n \le \max\left(f\left(A_s^n\right)\right) \vee \\ E_s^n \ge 2\pi - \max\left(f\left(A_s^n + \pi\right)\right) & \end{cases} \tag{5b}$$

where E_p^n and A_p^n represent the elevation and azimuth in the n-frame, respectively, p is considered as the center point of the sphere. Attention should be paid to the fact that the azimuth angle here may exceed π, as it may exceed the top and reach the other side. The function represents the functional relationship of the 3D-NLOS-DL, determining whether the signal from the sth satellite is LOS or NLOS. E_s^n and A_s^n represent the elevation and azimuth of the sth satellite.

3) 3-DMAP-Assisted GNSS Three-Step Positioning Algorithm: The algorithm flow is illustrated in Algorithm 1. First, all signals are used to calculate the initial positioning result, which includes NLOS signals. The number of satellites to retain is set to limit (with a minimum requirement of 4, ensuring $\text{limit}_1 \ge 4$). Then, excess NLOS signals are eliminated. Finally, filtering is performed based on time series change detection, with the specific detection criteria outlined in Algorithm 1 step C.

Algorithm 1 Three-Step-GNSS Positioning Algorithm

Init: Acquire satellite pseudo-range signals $\rho_{s,t}^{G,e}$ and 3D MAP data at time t.
Step A
1. Calculate the initial position $\mathbf{P}_1$ using all pseudo-range;
2. Calculate the 3D-NLOS-DL based on the 3D MAP;
Step B
3. If $Num_{satellite} > \text{limit}_1$
4. Eliminate excess NLOS satellite signals;
5. End
6. Calculate $\mathbf{P}_2$ using the remaining signals;
Step C
7. If $||\Delta\mathbf{P}_t|| > \text{limit}_2$
8. $\lambda = (\text{limit}_2/||\Delta\mathbf{P}_t||)^3$;
9. $\mathbf{P}_t = \mathbf{P}_{t-1} + \lambda\Delta\mathbf{P}_t$;
10. End
11. $\mathbf{P}_3 = \mathbf{P}_t$
Output: Three-Step-GNSS positioning results.

D. UWB Positioning Model Factor

UWB can be used for high-precision indoor or regional positioning. In the GNSS/UWB/IMU/MAP combined positioning framework described in this article, UWB primarily enhances indoor positioning and transition zone localization. The high-precision positioning factor in the UWB model involve three main issues.

1) The optimal deployment of anchors for a defined space.
2) The identification and removal of NLOS signals.
3) The high-precision calculation model based on data.

1) Optimal Deployment of Anchors: Optimal anchor deployment varies by spatial configuration, as incorrect configurations can cause significant positioning errors. The DOP value is typically used for optimization. Cao et al. [51] introduced weighted PDOP and optimizes configurations with the gray wolf algorithm, showing that the V-shape configuration provides the best average WPDOP at the tunnel's end.

In practical applications, more complex terrains may be encountered, and different spatial regions may have varying positioning accuracy requirements. To address this, we extend the WPDOP model by introducing the concept of SWPDOP and propose a gradient descent optimization algorithm for the optimal deployment of n anchors in any spatial domain. The SWPDOP value for a given spatial domain is expressed in (6) [51]

$$\text{SWPDOP} = \sum_{i=1}^{n} g(\mathbf{P}_i)\sqrt{\text{trace}\left(\boldsymbol{H}_i^T \boldsymbol{W} \boldsymbol{H}_i\right)^{-1}}/n \tag{6a}$$

$$\boldsymbol{W} = \text{diag}\left(1/\sigma_1^2, 1/\sigma_2^2, \ldots, 1/\sigma_m^2\right) \tag{6b}$$

$$\boldsymbol{H}_i = \begin{bmatrix} \boldsymbol{h}_1 & \boldsymbol{h}_2 & \cdots & \boldsymbol{h}_m \end{bmatrix}^T \tag{6c}$$

$$\boldsymbol{h}_m = \begin{bmatrix} (x - x_m)/d_m & (y - y_m)/d_m & (z - z_m)/d_m \end{bmatrix} \tag{6d}$$

where $g(\mathbf{P}_i)$ denotes the weights corresponding to different spatial locations, $\boldsymbol{H}_i$ represents the position matrix at point $\mathbf{P}_i$, $\boldsymbol{W}$ indicates the weight matrix of distance measurements,

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Algorithm 2 Gradient Descent Optimization Deployment Algorithm for UWB Anchors

Init: Obtain a spatial gridding model and determine the number of anchors (n).

1. Set initial coordinates for the anchors P_1 to P_n (either randomly or based on prior knowledge);
2. Calculate *SWPDOP*;
3. While $i < \text{limit}_3 \wedge j < \text{limit}_4$
4. $k = j\%n + 1$;
5. Generate a random position change vector $\Delta\mathbf{P}_k$;
6. Restricted area;
7. Calculate the updated $SWPDOP_k$;
8. If $SWPDOP_k < SWPDOP_{k-1}$;
9. Update $P_{1k} \ldots P_{nk}$;
10. $i++$;
11. End
12. $j++$;
13. End

Output: Optimized coordinate deployment plan.

σ_m signifies the noise standard deviation at the mth anchor point, (x, y, z) denotes the tag coordinates, and (x_m, y_m, z_m) represents the coordinates of the mth anchor point.

The gradient descent optimization deployment algorithm for anchor coordinates in any spatial domain is outlined in Algorithm 2. Initially, the anchor coordinates can be set based on empirical knowledge. A good initial value can significantly reduce the iteration optimization time. The algorithm iteratively calculates the *SWPDOP*, updating the anchor coordinates when a reduction is observed. To accelerate the update rate, only one coordinate is updated at a time, making it easier to meet the update conditions. This algorithm is considerably simpler than the gray wolf optimization algorithm, yet it proves to be highly effective.

2) NLOS Identification and Removal Problem: In UWB positioning, NLOS significantly affects ranging errors, leading to considerable inaccuracies in subsequent positioning results, necessitating their removal. Currently, no unified method exists. We developed a joint UWB NLOS signal identification approach that combines RSSI assessment [43] and anchor information communication, as expressed in

$$l_U = \begin{cases} 0|_{\text{NLOS}}, & E_1 \cup E_2 \\ 1|_{\text{LOS}}, & \neg E_1 \cap \neg E_2 \end{cases} \tag{7a}$$

$$E_1 : RSSI_{tr} - RSSI_{fp} > \text{limit}_5 \tag{7b}$$

$$E_2 : \text{Area}_A \neq \text{Area}_{\text{now}} \tag{7c}$$

where l_U indicates whether the signal from the Uth anchor is LOS or NLOS. Events E_1 and E_2 represent different NLOS judgment conditions. If any of these conditions are met, the signal is considered NLOS. $RSSI_{tr}$, $RSSI_{fp}$ denote the RSSIs of the total received signal and the first path signal, respectively. Limit_5 represent constant threshold values. Area_A and Area_{now} represent the region where the Uth anchor is located and the region where the current tag is located.

3) Positioning Solution Model: Based on the different measurement signals, UWB positioning calculations can be categorized into three modes: TOA, TDOA, and AOA, with TOA and TDOA being more commonly used. This article employs the TOA ranging calculation model.

Ranging methods include time of flight (TOF) and two-way ranging (TWR), where TOF requires system time synchronization while TWR does not. After obtaining ranging information, prefiltering and NLOS detection are conducted before the positioning calculation. Numerous scholars have proposed high-precision distance positioning algorithms from various perspectives. Considering application constraints, the two-step maximum likelihood (TSML) algorithm can achieve the Cramer–Rao lower bound (CRLB) and is therefore typically regarded as optimal [51], [52].

Additionally, since the TSML model requires at least four anchors to calculate tag positions, a supplementary explanation for cases with fewer than four accessible anchors is provided in

$$\begin{bmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{bmatrix} = \begin{cases} \begin{bmatrix} \sqrt{|\omega_1|} & \sqrt{|\omega_2|} & \sqrt{|\omega_3|} \end{bmatrix}^T, & N \geq 4 \\ \begin{bmatrix} \sqrt{|\omega_1|} & \sqrt{|\omega_2|} & \sum_{i=1}^{3} z_i/3 \end{bmatrix}^T, & N = 3 \\ \sum_{i=1}^{2} \begin{bmatrix} x_i & y_i & z_i \end{bmatrix}^T/2, & N = 2 \\ \begin{bmatrix} x_1 & y_1 & z_1 \end{bmatrix}^T, & N = 1 \end{cases} \tag{8}$$

where N represents the number of anchors that the tag can detect. When the number of anchors is greater than or equal to 4, (x, y, z) is calculated by 3-D TSML model, and the $\omega = \begin{bmatrix} \omega_1 & \omega_2 & \omega_3 \end{bmatrix} = \begin{bmatrix} \hat{x}^2 & \hat{y}^2 & \hat{z}^2 \end{bmatrix}$ is the result variable of the TSML algorithm. With 3 anchors, (x, y) is calculated using the 2-D TSML model, while z is estimated using the average of the anchors. With 2 anchors, position estimation is done using the anchor average, and with 1 anchor, the anchor coordinate is directly used for estimation.

Finally, a combination of time series change detection filtering, Kalman filtering, and mean filtering is applied to enhance positioning accuracy and stability.

E. Quality Assessment Adaptive Fusion Factor

In a combined navigation system, the performance of each subsystem varies with multiple factors, making the efficient and accurate fusion of positioning results crucial. A prevailing approach is to adjust the fusion ratios of subsystems adaptively through quality assessment [53], [54], ensuring the accuracy of the fused system. The GNSS/UWB/IMU/MAP model in this study also employs this strategy. For each subsystem, the quality assessment model can be formulated as a function of relevant state parameters, and through normalization, the fusion ratios can be obtained, as shown in

$$\alpha_t^G = f^G\left(S_{\text{number}}, \text{dis}_t^G, \boldsymbol{e}_{\text{map}}^G, \Delta\mathbf{P}_t^G\right) \tag{9a}$$

$$\alpha_t^U = f^U\left(A_{\text{number}}, \text{dis}_t^U, \boldsymbol{e}_{\text{map}}^U, \Delta\mathbf{P}_t^U\right) \tag{9b}$$

$$\alpha_t^{INS} = f^{INS}\left(\text{dis}_t^{INS}, \boldsymbol{e}_{\text{map}}^{INS}, \Delta\mathbf{P}_t^{INS}\right) \tag{9c}$$

$$\alpha_t^{PDR} = f^{PDR}\left(\text{dis}_t^{PDR}, \boldsymbol{e}_{\text{map}}^{PDR}, \Delta\mathbf{P}_t^{PDR}\right) \tag{9d}$$

$$\boldsymbol{\alpha}_t = \begin{bmatrix} \alpha_t^G & \alpha_t^U & \alpha_t^{INS} & \alpha_t^{PDR} \end{bmatrix} \tag{9e}$$

$$\boldsymbol{Q}_t = \boldsymbol{\alpha}_t / \left(\alpha_t^G + \alpha_t^U + \alpha_t^{INS} + \alpha_t^{PDR}\right) \tag{9f}$$

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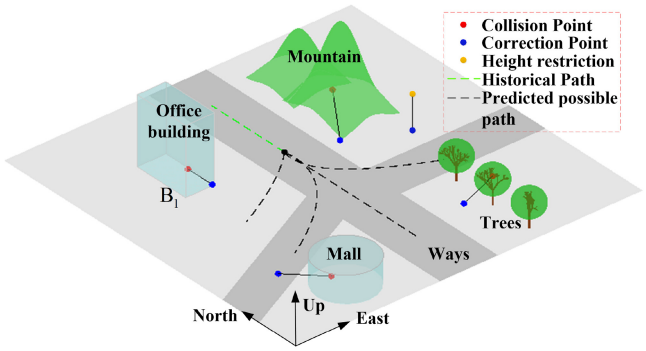


Fig. 6. MAP constraints and path prediction factors.

where α_t^G, α_t^U, α_t^{INS}, and α_t^{PDR} represent the quality evaluation estimates of the four subsystems: GNSS, UWB, INS, and PDR. Their value range is [0,1], and the quality assessment functions f^G, f^U, f^{INS}, and f^{PDR} are used for calculation. S_{number} denotes the number of available GNSS satellites, and A_{number} denotes the number of available UWB anchors. dis_t^G, dis_t^U, dis_t^{INS}, and dis_t^{PDR} represent the dispersion of the positions of the four subsystems at the current moment, and their values are inversely proportional to the distance from the average position. $\boldsymbol{e}_{\text{map}}^G$, $\boldsymbol{e}_{\text{map}}^U$, $\boldsymbol{e}_{\text{map}}^{INS}$, and $\boldsymbol{e}_{\text{map}}^{PDR}$ represent the errors between the subsystems and the map correction prediction system. $\Delta\mathbf{P}_t^G$, $\Delta\mathbf{P}_t^U$, $\Delta\mathbf{P}_t^{INS}$, and $\Delta\mathbf{P}_t^{PDR}$ represent the positional differences between the subsystems at the current moment and the previous moment. $\boldsymbol{Q}_t$ is the fusion ratio data at time t.

F. MAP Constraints and Path Prediction Factor

The application of electronic map models has become increasingly widespread, serving as an important auxiliary correction method for positioning, such as lane constraint assistance in vehicle navigation. Currently, the main challenge is the acquisition of accurate 3-D maps. However, with advancements in automated surveying instruments like LiDAR, precise 3-D map surveying will no longer be difficult. It is anticipated that research on positioning technologies assisted by 3-D electronic maps will experience rapid growth.

MAP constraints and path prediction factors can be divided into constraint factors and path prediction factors, as illustrated in Fig. 6. Constraint factors correct the position through collision detection between positioning points and the 3-D map, as well as height constraints. Path prediction factors are designed and implemented using conditionally extended nonstationary Markov models.

1) 3-D MAP Constraint Factors: The fused position at the current moment is obtained based on the quality fusion factor and converted to the 3-D map model. This position is then compared with the structural objects within the map and corrected through collision constraints and height constraints. For pedestrians, their position must not be embedded within object models, and their height is within a certain range above the ground. Based on these constraints, the 3-D MAP constraint factors are established as shown in

$$\boldsymbol{V}_{\text{res}} = \begin{cases} k\left(\mathbf{P}_{\text{col}}^n - \mathbf{P}_t^n\right), & \mathbf{P}_t^n \in \left\{\boldsymbol{B}_i^n, \boldsymbol{W}_j^n, \boldsymbol{T}_k^n, \boldsymbol{G}_l^n\right\} \\ \mathbf{0}_{3\times 1}, & \mathbf{P}_t^n \notin \left\{\boldsymbol{B}_i^n, \boldsymbol{W}_j^n, \boldsymbol{T}_k^n, \boldsymbol{G}_l^n\right\} \end{cases} \tag{10a}$$

$$\mathbf{P}_{t,\text{res}}^n = \mathbf{P}_t^n + \boldsymbol{V}_{\text{res}} \tag{10b}$$

$$\mathbf{P}_{t,\text{res}}^n(z) = f^{\text{MAP}}(x, y) + h \tag{10c}$$

where $\boldsymbol{V}_{\text{res}}$ represents the collision correction term. $\mathbf{P}_{\text{col}}^n$ represents the external point of the landmark where the probability of collision is the highest with the current position $\mathbf{P}_t^n$, and k is the proportional coefficient. $f^{\text{MAP}}(x, y)$ represents the altitude of the map model at the location (x, y). h is the pedestrian altitude correction, which corrects the altitude by matching the corresponding altitude value for the horizontal position.

2) 3-D MAP Path Prediction Factors: Based on historical trajectories, map landmarks, road distribution, and pedestrian intentions, the future path of pedestrians can be predicted. If the prediction is accurate, it will yield high-precision predicted coordinates that can be used to correct the current pedestrian position, significantly enhancing positioning accuracy and robustness. The path prediction for pedestrians can be addressed using conditionally extended nonstationary Markov models, with the specific prediction model introduced below.

First, establish the state set as shown in

$$\boldsymbol{S}\left(\mathbf{P}_t^n, t, con\right) = \left\{\boldsymbol{B}_i^n\left(\mathbf{P}_t^n, t, con\right), \ldots, \boldsymbol{W}_j^n\left(\mathbf{P}_t^n, t, con\right), \ldots \right. \\ \left. \boldsymbol{T}_k^n\left(\mathbf{P}_t^n, t, con\right), \ldots, \boldsymbol{G}_l^n\left(\mathbf{P}_t^n, t, con\right)\right\} \tag{11}$$

where $\boldsymbol{S}$ represents the set of states, which changes with time t and position $\mathbf{P}_t^n$. $\boldsymbol{B}_i^n$,$\boldsymbol{W}_j^n$,$\boldsymbol{T}_k^n$,$\boldsymbol{G}_l^n$ represent the specific states which are the current positions of the pedestrian. These states are observable, but over time, spatial $\mathbf{P}_t^n$ and condition con change, thus forming the conditionally expanded nonstationary Markov model.

The initial state is directly obtained from the positioning model observations. Due to the influence of errors, the probability of any observed initial state $\boldsymbol{X}(0) = S(\mathbf{P}_0^n, 0, con)$ is determined by the distance between the positioning coordinate point and the surrounding map model objects, as well as parameters, such as object type.

The expression for the probability matrix of conditional state transitions is shown in

$$\mathbb{P}_{n\times n,t} = \left\{\mathbb{P}_{ij} \mid i, j = 1, \ldots, n\right\}, n = \text{length}(\boldsymbol{S}) \tag{12a}$$

$$\mathbb{P}_{ij,t} = \mathbb{P}(s_t \mid s_{t-1}, \boldsymbol{X}(t)) = \xi\left(\mathbf{P}_t^n, t, con\right) \tag{12b}$$

where $\mathbb{P}_{n\times n,t}$ represents the transition probability matrix from n states at time $t-1$ to n states at time t. $\mathbb{P}_{ij,t}$ is the probability element, calculated through the conditional probability function. ξ denotes the conditional probability distribution function corresponding to the specific state.

The state probability at time t is shown in

$$\boldsymbol{\gamma}_{1\times n,t} = \{\gamma_i \mid i = 1, \ldots, n\}, n = \text{length}(\boldsymbol{S}) \tag{13a}$$

$$\gamma_{i,t} = \sum_{j=1}^{n} \mathbb{P}\left(s_{i,t} \mid s_{j,t-1}, \boldsymbol{X}(t)\right) \tag{13b}$$

where $\boldsymbol{\gamma}_{1\times n,t}$ represents the probability of the occurrence of the n states at time t, and their sum is 1.

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By continuous state transition changes, the maximum probability state and its transition path within k steps can be obtained, as shown in

$$s_{\max}^{t+k} = \operatorname{argmax} \boldsymbol{\gamma}_{t+k} = \operatorname{argmax} \boldsymbol{\gamma}_t \prod_{i=1}^{k} \mathbb{P}_{t+i} \quad (14a)$$

$$\mathbf{Path}_{\max}^{k} = \left\{ s^t, s_{\max}^{t+1}, \ldots, s_{\max}^{t+k} \right\} \quad (14b)$$

where $s_{\max}^{t+k}$ represents the state with the highest probability at the kth step, and $\mathbf{Path}_{\max}^{k}$ represents the set of sequences of states that have the highest probability over k steps.

By calculating predicted state probabilities from historical data and integrating them with observed state probabilities, the maximum probability location serves as a correction value for the current position, leading to a more accurate corrected position.

G. Joint Filter Factor

After integrating MAP data, the positioning results achieve high accuracy. To further improve system smoothness and robustness, a filtering algorithm is necessary to handle errors and sudden changes in the trajectory.

When processing the position sequence with MAP data, short-term mismatches may occur, causing abrupt trajectories and misaligned points. These mismatches, which result in sudden changes persisting for a while before returning to normal, cannot be fully addressed by standard filters. A spike detection filter is thus designed to handle such cases. For single-point mismatches, they are treated as high-frequency noise and removed using a low-pass filter. The Joint Filter process is detailed in Algorithm 3.

Algorithm 3 Joint Filter

Init: Obtain the three-dimensional position coordinate sequence $\mathbf{P} = [\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}]$ corrected by MAP data.

1. Calculate the differential value $d\boldsymbol{x} = \boldsymbol{x}(k) - \boldsymbol{x}(k-1)$;
2. $d\boldsymbol{x}(d\boldsymbol{x} < \text{limit}_6) = 0$;
3. $a = find(d\boldsymbol{x} \neq 0)$;
4. $b = a$;
5. For i = 2:length(a) − 1
6. If $a(i) - a(i-1) < \text{limit}_7 \wedge a(i+1) - a(i) < \text{limit}_7$
7. $b(i) = 0$;
8. End
9. If $a(i) - a(i-1) > \text{limit}_7 \wedge a(i+1) - a(i) > \text{limit}_7$
10. $b(i) = 0$;
11. End
12. End
13. $a = b(b \neq 0)$;
14. for $i = 1$: length(a)/2
15. $t = 2i - 1$;
16. Perform linear interpolation between $\boldsymbol{x}(a(t) - 1)$ and $\boldsymbol{x}(a(t+1) + 1)$.
17. End
18. $\boldsymbol{x} = Low_Pass_Filter(\boldsymbol{x})$;
19. Repeat steps 1 to 18 for $\boldsymbol{y}$ and $\boldsymbol{z}$.

Output: The position coordinate after joint filtering.

IV. Experimental Results

To validate the models and algorithms in this article and analyze performance, this section designs comprehensive evaluation experiments. Section IV-A introduces the equipment, scenarios, and metrics. Section IV-B verifies the NLOS elimination model and GNSS three-step positioning method. Section IV-C assesses the UWB deployment algorithm. Section IV-D evaluates seamless indoor and outdoor positioning experiments. Finally, Section IV-E comprehensively discusses the experimental results.

A. Experimental Setup and Metrics

The experimental equipment includes smartphones, a UWB system, and a reference system, as shown in Fig. 7(a). The smartphones used for testing are the Huawei HONOR V20 and iQOO U3. The UWB system employs LinkTrack P-A model from the company Nooploop. The reference system utilizes RTK and 3-D MAP to obtain the true position trajectory. The RTK device is the WTRTK-M model from WiT Intelligent Company, paired with a four-satellite full-band antenna.

The experimental scenarios, depicted in Fig. 7(b), include various rich testing environments: A. Terrain and tree obstructions, B. Building obstructions, C. Outdoor stairs, D. Indoor lifts, E. Indoor corridors, F. Indoor stairs, and G. Overpass obstructions. These scenarios comprehensively test the algorithms presented in this article. The colored lines in the figure represent the running trajectory, with the sequence being A-D-E-F-G-B-C-A, marked on 2-D satellite map.

The experimental metrics include the root-mean-square error (RMSE), mean error (MEAN), standard deviation (STD), and maximum error (MAX) in the ENU coordinate system [49]. Additionally, trajectory plots, signal quantity variation graphs, error variation graphs, and error cumulative distribution function (CDF) plots will be created to achieve a comprehensive evaluation of the systems and algorithms presented in this article.

Additionally, to verify the effectiveness of our proposed method, we compared them with the following baseline algorithms in part D.

1) *GPS-UWB-PDR-CV [49]:* The GNSS tightly coupled (TC) Doppler/pseudorange factor in the literature was replaced with the constant velocity (CV) factor, and when UWB is available, the UWB CV factor is incorporated to adapt to the devices used in this article.
2) *GPS-UWB-PDR-CV-SMM [49]:* The smoothness-driven motion model (SMM) factor was incorporated based on the GPS-UWB-PDR-CV framework.
3) *GPS-UWB-INS-EKF [55]:* Use EKF as the subfilters, and use federal filter for main filter.
4) *Fusion-Map (Proposed):* GPS/INS/PDR/UWB Fusion system improved by MAP Constraints and Prediction.
5) *Fusion-Map-JF (Proposed):* Fusion-Map system improved by Joint Filter.

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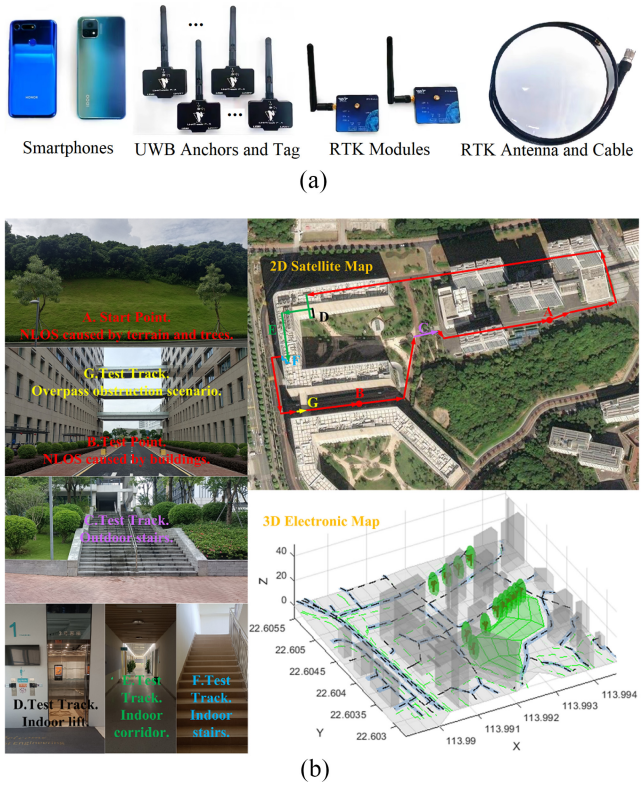


Fig. 7. Experimental equipment and scene setup. (a) Main experimental equipment and reference equipment. (b) Experimental scenarios and trajectories.

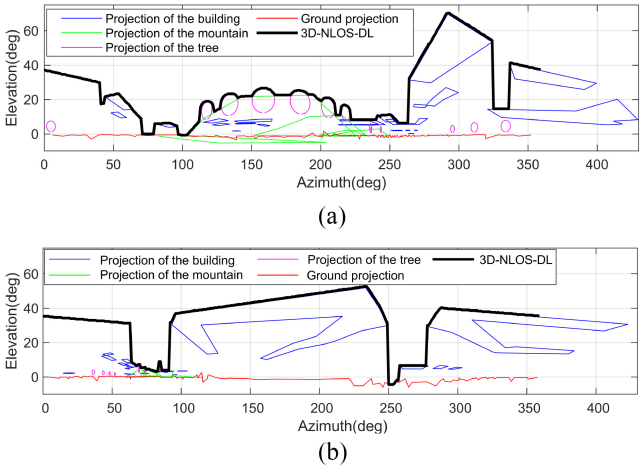


Fig. 8. Comparison of NLOS decision lines in different scenes. (a) 3D-NLOS-DL of Scene A. (b) 3D-NLOS-DL of Scene B.

B. Validation of the GNSS Three-Step Positioning Algorithm

First, the GNSS model presented in this article is validated, focusing on improving positioning performance in NLOS environments. The experimental scenarios are at points A and B shown in Fig. 7(b). Point A primarily includes obstructions from hilly terrain, trees, and buildings, while point B mainly features building obstructions and bridge obstructions. The 3D-NLOS-DL for these two different locations is illustrated in Fig. 8.

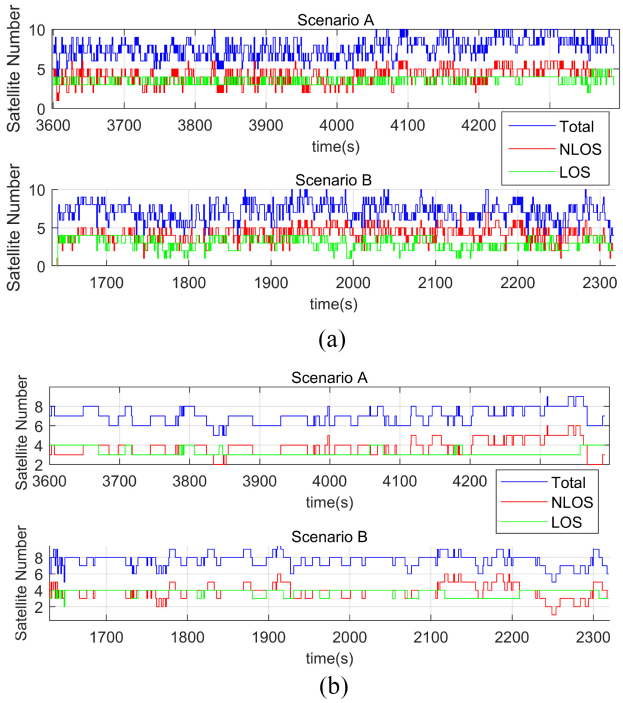


Fig. 9. Variation in the number of satellites detected by different phones. (a) Satellite number detected by a Honor phone. (b) Satellite number detected by an iQOO phone.

Fig. 8(a) corresponds to scene A, where the projections of the obstructing objects at various azimuths are displayed in conjunction with Fig. 7(b). In the figure, the blue line represents the projection of building tops, the pink line represents tree projections, the green line indicates the projection of hills, the red line denotes the terrain surface projection, and the bold black line represents the 3D-NLOS-DL. Fig. 8(b) corresponds to scene B, where the projections of two buildings dominate, serving as the main cause of NLOS. The 3D-NLOS-DL shown in Fig. 8 allows for quick and convenient determination of NLOS satellites at any given moment for that point.

Data from GPS satellites are collected for a specified duration in both scenes, and NLOS determination is conducted using the 3D-NLOS-DL shown in Fig. 8. The variations in the number of visible satellites, NLOS satellites, and LOS satellites are illustrated in Fig. 9.

It can be observed that after starting the positioning process, the number of GPS satellites detected increases rapidly and stabilizes around 7 to 8 satellites. The Honor phone detects slightly more satellites, but the data exhibits greater fluctuations, while the iQOO phone shows the opposite trend. The impact of both scenarios on the number of NLOS satellites is quite similar, averaging around 3–4 satellites, which constitutes approximately 43%–50% of the total.

Using the three-step positioning algorithm proposed in this article, data processing is performed by removing NLOS satellites based on the required number of satellites and then applying time series spike detection for filtering. The results from the proposed algorithm are compared with the direct positioning results, as shown in Fig. 10 (the map has been shifted downward by 50 m for clarity).

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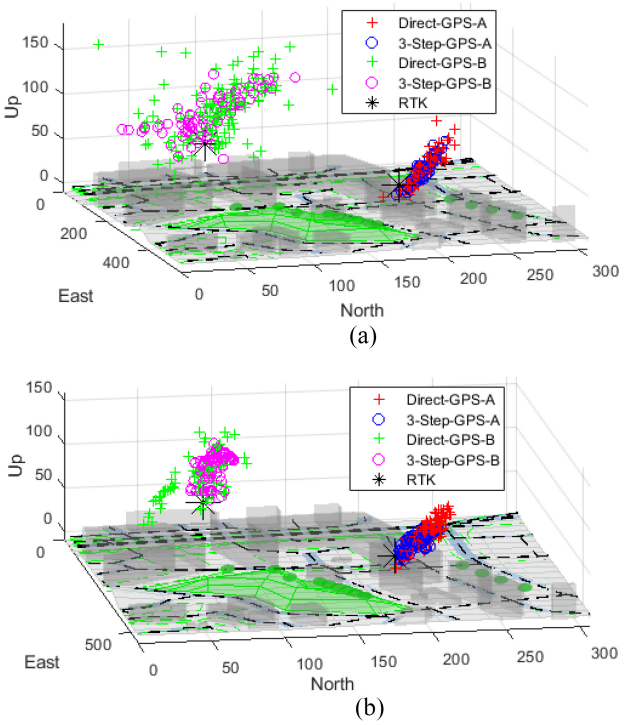


Fig. 10. Variation in the number of satellites detected by different phones. (a) Comparison of positioning results from a Honor phone. (b) Comparison of positioning results from an iQOO phone.

TABLE I
COMPARISON OF GPS ALGORITHMS ACROSS DIFFERENT DEVICES

Scenes	Algorithms	RMSE (m)	MEAN (m)	STD (m)	MAX (m)
Device: Honor					
A	Direct-GPS	27.72	26.74	8.34	78.40
	Three-Step-GPS	**24.33**	**23.78**	**6.48**	**60.85**
B	Direct-GPS	62.50	62.49	28.32	635.03
	Three-Step-GPS	**44.67**	**38.38**	**18.78**	**111.54**
Device: iQOO					
A	Direct-GPS	30.38	29.88	8.903	68.16
	Three-Step-GPS	**20.60**	**20.30**	**5.834**	**48.39**
B	Direct-GPS	**38.80**	**38.80**	14.84	121.16
	Three-Step-GPS	46.54	46.20	**8.60**	**77.45**

It can be observed that at point A, both smartphones show a high degree of clustering in the direct positioning results and the results from the proposed algorithm. However, the clustering centers exhibit some deviation from the RTK results. After applying the proposed algorithm, the large deviation data significantly decreases, reducing both the deviation error and dispersion. For point B, the positioning results are more scattered and exhibit larger errors, primarily due to the smaller visible space, which leads to unstable satellite signal access. The proposed algorithm enhances the clustering of the positioning results, greatly improving stability and reducing deviation errors to some extent.

The performance metrics of the experimental results are compared in Table I. It can be observed that, except for the iQOO phone's positioning results in scene B, both RMSE and MEAN show significant decreases, demonstrating the effectiveness of the proposed algorithm in enhancing accuracy.

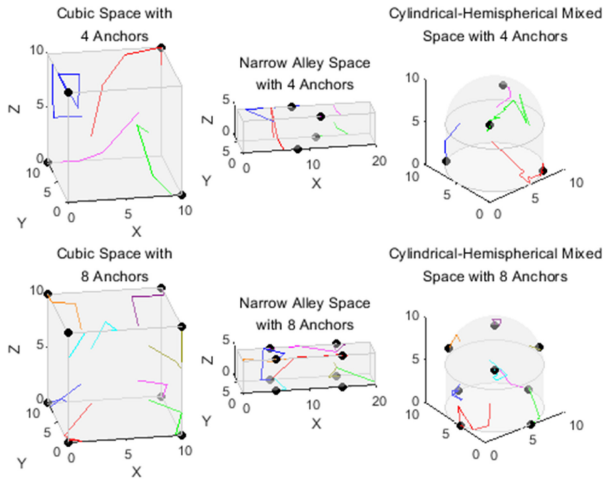


Fig. 11. Optimization results of anchor deployment in different spaces.

For STD and MAX, the use of the three-step positioning method also results in substantial reductions, confirming the effectiveness of the proposed algorithm in improving stability.

C. Validation of UWB Optimization Deployment Algorithm

To validate the performance of the UWB anchor deployment algorithm proposed in this article, comparative experiments with different spatial shapes, spatial weighting functions, and numbers of UWB anchors were designed. The variations for each scenarios are presented to demonstrate the effectiveness of the proposed algorithm.

In the case of unweighted spatial configurations, the optimization comparisons for different spatial shapes and numbers of anchors are shown in Fig. 11. It illustrates the optimized deployment results for 4 anchors and 8 anchors in different spatial domains, where the colored lines represent the movement trajectories of the anchors, and the black dots indicate the final positions. For the 4-anchor configuration, the optimization results yield a tetrahedral shape, maximizing the size, while for the 8-anchor configuration, the results yield either a cubic or staggered cubic shape.

The functions are set as W_{s1}, W_{s2}, and W_{s3} for the cubic, narrow corridor, and cylindrical-hemisphere configurations, respectively, as shown in

$$W_{s1} = -\left(r/10\sqrt{(3)}\right)^{0.1} + 1, 0 \le r \le 10\sqrt{3} \quad (15a)$$

$$W_{s2} = -(x/20)^{0.1} + 1, 0 \le x \le 20 \quad (15b)$$

$$W_{s3} = -(z/10)^{0.1} + 1, 0 \le z \le 10 \quad (15c)$$

where r represents the radial distance in polar coordinates, while x and z denote the coordinates in the Cartesian coordinate system.

The optimization comparison results after setting the spatial weighting functions are shown in Fig. 12. It can be observed that after weighting, the impact on the corridor is significant. For the narrow alley space, although the configuration remains unchanged, it has clearly shifted to the left. This is related to the weighting functions, with a larger weight on the left side indicating higher precision requirements in that direction, and

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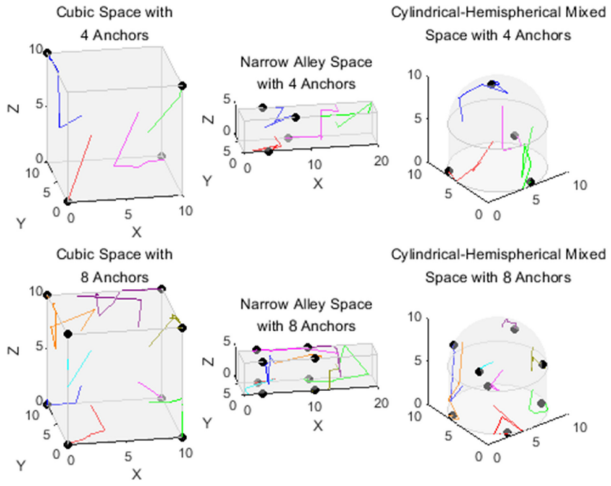


Fig. 12. Optimization results of spatially weighted anchor deployment.

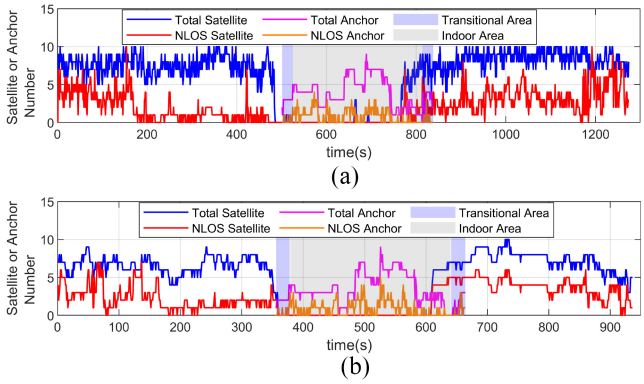


Fig. 13. Variation of satellite count and UWB anchor count in different scenarios with different mobile phones. (a) Experimental results of honor smartphone. (b) Experimental results of iQOO smartphone.

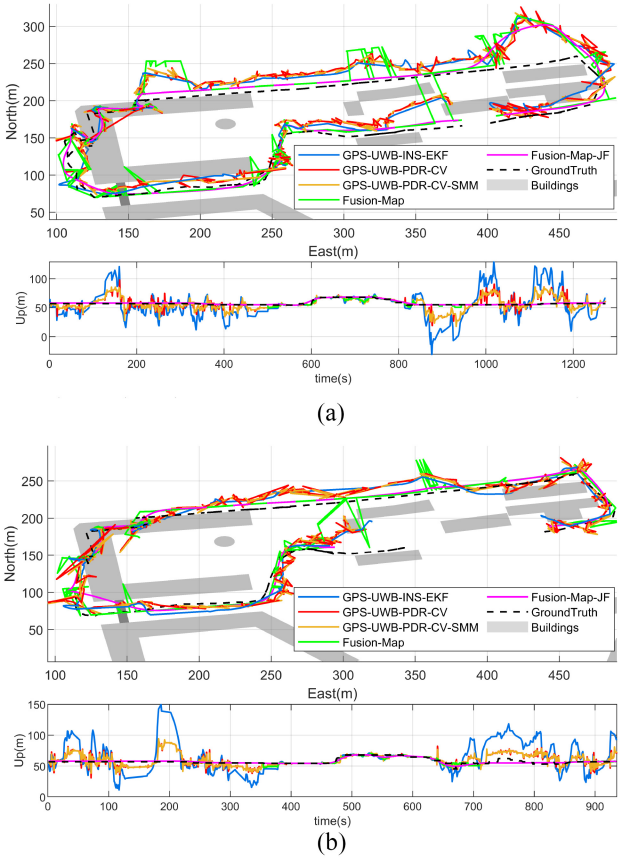


Fig. 14. Comparison of positioning trajectories of different mobile phones at different locations. (a) Indoor and outdoor positioning trajectories of honor smartphone. (b) Indoor and outdoor positioning trajectories of iQOO smartphone.

the optimization results align with this requirement. The cube and cylindrical-hemispherical configurations with 4 anchors are similarly affected, with the weighting functions demanding high precision at the origin and the bottom, leading to an angular rotation of the configurations. As a result, the cube's origin includes 1 anchor, while the cylindrical-hemispherical bottom contains 3 anchors, further demonstrating the effectiveness.

D. Comprehensive Experimental Evaluation of Seamless Indoor and Outdoor Positioning

To thoroughly validate the proposed model architecture, continuous positioning experiments in various scenarios (Fig. 7(b)) were conducted. Fig. 13 illustrates the variations in the total number of satellites and UWB anchors along the experimental trajectory.

The Honor phone shows more frequent fluctuations in GPS satellite counts and detects slightly more satellites compared to the iQOO phone, which exhibits stronger local stability. However, both phones display largely consistent global trajectories, confirming that NLOS recognition remains stable in similar environments and validating the NLOS identification algorithm proposed.

UWB signals are available in transitional areas, though in lower quantities. Indoors, the number of UWB anchors varies with deployment, peaking at 9, though some signals are NLOS. The consistent count variations across both experiments align with NLOS patterns, confirming the stability of NLOS characteristics in similar environments. This suggests that NLOS identification can be addressed as a prior-based matching problem depending on the environment.

The positioning results of the two smartphones in complex indoor and outdoor environments are shown in Fig. 14(a) and (b). The gray area in the upper left represents indoor positioning areas, while the rest covers outdoor and transitional areas. The performance of different algorithms is consistent across both smartphones, confirming their cross-platform applicability. GPS-UWB-INS-EKF shows good planar accuracy and smoothness but has significant height error fluctuations. GPS-UWB-PDR-CV and GPS-UWB-PDR-CV-SMM yield similar results, with much better height accuracy than EKF. The two algorithms proposed in this article both achieve high height accuracy, relying on the effectiveness of (10). Fusion-Map has some mismatches, such as the green protrusion at the top of Fig. 14(a), but overall accuracy is high. Fusion-Map-JF, using joint filtering, corrects most errors and smooths the results, achieving the best performance in

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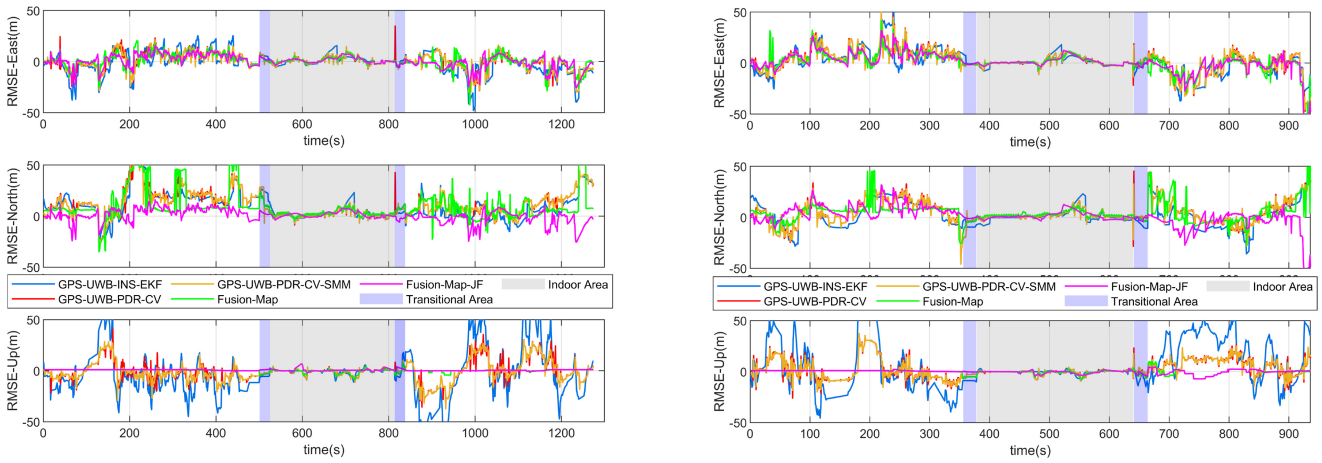


Fig. 15. Comparison of RMSEs for different mobile phones and algorithms. (a) Comparison of RMSEs for different algorithms on honor smartphone. (b) Comparison of RMSEs for different algorithms on iQOO smartphone.

TABLE II
COMPARISON OF POSITIONING METRICS FOR DIFFERENT ALGORITHMS IN OUTDOOR SCENARIOS

Device	**Honor V20**				**iQOO U3**			
Algorithms	RMSE (m)	MEAN (m)	STD (m)	MAX (m)	RMSE (m)	MEAN (m)	STD (m)	MAX (m)
GPS-UWB-INS-EKF [55]	26.36	12.41	13.82	91.57	28.79	10.09	16.66	94.41
GPS-UWB-PDR-CV [49]	19.72	14.82	8.70	79.06	18.07	7.83	9.78	62.47
GPS-UWB-PDR-CV-SMM [49]	19.47	14.69	8.55	71.70	17.96	7.77	9.74	64.27
Fusion-Map (proposed)	14.06	11.19	7.31	74.15	12.22	5.72	6.26	77.75
Fusion-Map-JF (proposed)	**10.83**	**7.49**	**4.80**	**55.76**	**10.77**	**4.76**	**5.50**	**53.23**

TABLE III
COMPARISON OF POSITIONING METRICS FOR DIFFERENT ALGORITHMS IN TRANSITIONAL SCENARIOS

Device	**Honor V20**				**iQOO U3**			
Algorithms	RMSE (m)	MEAN (m)	STD (m)	MAX (m)	RMSE (m)	MEAN (m)	STD (m)	MAX (m)
GPS-UWB-INS-EKF [55]	14.71	13.14	6.53	27.59	11.46	10.88	2.89	18.26
GPS-UWB-PDR-CV [49]	12.24	11.61	5.38	55.70	6.09	**2.58**	3.41	54.38
GPS-UWB-PDR-CV-SMM [49]	11.29	10.50	4.77	29.65	5.92	2.80	3.34	41.20
Fusion-Map (proposed)	11.56	10.94	3.01	25.31	5.95	5.44	**1.86**	16.56
Fusion-Map-JF (proposed)	**10.77**	**10.38**	**2.54**	**17.45**	**4.76**	4.30	1.92	**12.84**

TABLE IV
COMPARISON OF POSITIONING METRICS FOR DIFFERENT ALGORITHMS IN INDOOR SCENARIOS

Device	**Honor V20**				**iQOO U3**			
Algorithms	RMSE (m)	MEAN (m)	STD (m)	MAX (m)	RMSE (m)	MEAN (m)	STD (m)	MAX (m)
GPS-UWB-INS-EKF [55]	4.42	2.48	2.77	25.41	4.83	**1.75**	3.02	23.90
GPS-UWB-PDR-CV [49]	4.05	**2.13**	2.38	15.66	4.12	2.22	2.57	36.41
GPS-UWB-PDR-CV-SMM [49]	4.12	**2.13**	2.42	15.65	4.15	2.24	2.58	26.56
Fusion-Map (proposed)	4.04	3.14	1.94	14.75	3.97	3.12	1.94	15.96
Fusion-Map-JF (proposed)	**3.79**	3.18	**1.76**	**10.96**	**3.91**	3.10	**1.87**	**13.81**

both smartphone experiments and demonstrating the superior accuracy and stability of the proposed algorithms.

The comparison of the three-axis RMSE is shown in Fig. 15, which clearly illustrates the changes in error over time and spatial location. For the experiments conducted with the two smartphones, the final results (in pink) demonstrate a significant suppressive effect on various fluctuations and deviations, indicating that their stability and accuracy are undoubtedly optimal. This also confirms the algorithm's applicability across different smartphone platforms.

Additionally, in the transitional areas, Fusion-Map-JF maintained optimal accuracy and stability without any sudden spikes, ensuring seamless positioning between indoor and outdoor environments.

Tables II–IV compare the performance metrics of the experimental results for outdoor, transitional, and indoor environments, respectively. From the outdoor results, it is clear that due to the poor baseline accuracy of consumer-grade sensors in standard smartphones, the EKF algorithm's error does not converge, resulting in large errors (26.36 and 28.79 m) and the worst stability, with STD values of 13.82 and 16.66 m. The GPS-UWB-PDR-CV and GPS-UWB-PDR-CV-SMM algorithms show significant improvements in both accuracy and stability compared to EKF, with the SMM-enhanced method

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performing better, achieving accuracies of 19.47 and 17.96 m. The Fusion-MAP and Fusion-MAP-JF algorithms proposed in this article offer substantial improvements in both accuracy and stability over existing algorithms. For both phones, the best performing Fusion-MAP-JF algorithm achieves accuracies of 10.83 and 10.77 m, representing improvements of 44.4% and 40.0% over GPS-UWB-PDR-CV-SMM. The stability values are 4.80 and 5.50 m, reflecting improvements of 43.9% and 43.5%, respectively.

In the transitional areas, due to the integration of high-precision UWB, most algorithms show significant improvements. The GPS-UWB-INS-EKF algorithm achieves accuracies of 14.71 and 11.46 m, while GPS-UWB-PDR-CV-SMM reaches 11.29 and 5.92 m. Fusion-MAP-JF further improves accuracy to 10.77 and 4.76 m, representing increases of 4.61% and 19.6% compared to GPS-UWB-PDR-CV-SMM. In terms of stability, Fusion-MAP-JF achieves 2.54 and 1.92 m, representing improvements of 46.8% and 42.5%.

For indoor areas, the accuracy and stability of the various algorithms are similar, but Fusion-MAP-JF still achieves the best results. Its accuracies are 3.79 and 3.91 m, improving by 8.00% and 5.78% compared to GPS-UWB-PDR-CV-SMM. In terms of stability, the values are 1.76 and 1.87 m, reflecting improvements of 27.3% and 27.5%.

Additionally, it can be observed that in the experiments, Fusion-MAP-JF shows the best improvement outdoors, while the enhancement in transitional and indoor environments is relatively lower. This is due to the poor baseline accuracy of the outdoor system, where map corrections play a significant role in improving accuracy and stability (by about 40%). In contrast, the baseline accuracy is higher in transitional and indoor settings, resulting in limited correction from the map and, a more modest performance improvement.

Furthermore, some MEAN values of Fusion-MAP-JF in transitional and indoor environments did not reach optimal levels. This is primarily due to a greater emphasis on minimizing STD. While lower STD results in smoother trajectories, it may also lead to a bias toward one side of the Ground Truth, thereby increasing the MEAN value.

The CDF is shown in Fig. 16. It can be observed that GPS-UWB-INS-EKF is the flattest, indicating its poor performance, while GPS-UWB-PDR-CV and GPS-UWB-PDR-CV-SMM show significant improvements over EKF, with similar performance. Fusion-MAP and Fusion-MAP-JF are the steepest, demonstrating the superior performance of the algorithms. Additionally, the results from both smartphones are consistent, confirming the algorithm's generalizability.

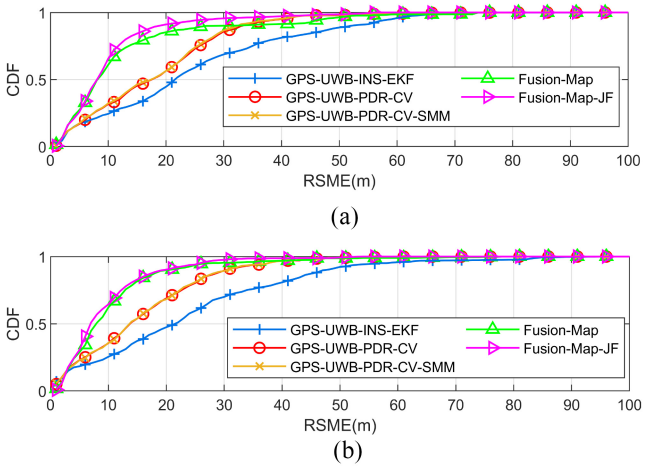


Fig. 16. Comparison of error CDF plots for different phones. (a) Error CDF plot for Honor phone. (b) Error CDF plot for iQOO phone.

E. Summary and Discussion of Experimental Results

In complex, continuously changing environments, this study achieved high-precision and high-stability seamless indoor-outdoor positioning results. This was done by integrating consumer-grade GPS, IMU, and UWB (which has already become an embedded sensor in some smartphones) with MAP data. Utilizing algorithms, such as Three-Step-GPS, INS, PDR, TSML-UWB, quality assessment fusion, MAP data correction, and fusion filtering. Experimental results demonstrate the advantages of the proposed algorithm in improving both accuracy and stability.

The improvements were not achieved overnight but were the result of multiple algorithms working collaboratively. The key lies in isolating the impact of errors at each stage, thus achieving consistent performance enhancement.

The Three-Step-GPS algorithm involves NLOS identification based on 3D-NLOS-DL and time series anomaly filtering. NLOS improves accuracy, while anomaly filtering enhances stability. Furthermore, NLOS identification should maintain a sufficient number of visible satellites to ensure the removed NLOS signals account for the dominant errors.

Both INS and PDR used fusion data for feedback correction, further improving local accuracy and stability on top of the Three-Step-GPS algorithm. Additionally, for consumer-grade IMUs in smartphones, PDR generally outperforms INS.

UWB can theoretically achieve centimeter-level accuracy, but in complex wide-area multipath environments, precision may degrade to the meter level. This study enhanced indoor positioning accuracy and stability by integrating UWB with INS, PDR, MAP data, and filtering algorithms.

MAP data was mainly used for collision correction and prediction. Collision correction involves adjusting the direction, with the maximum probability matching position serving as the correction direction to improve accuracy. Prediction helps align the position with the most probable path, although excessive errors may lead to incorrect path matching. Moreover, it transforms traditional positioning issues into matching problems, calculating the most likely matching positions on the map based on existing information. Theoretically, as long as the map accuracy and matching accuracy are sufficiently high, positioning accuracy can be infinitely improved, surpassing the limitations of traditional methods. This matching-based positioning approach is likely to become a focal point in future pedestrian localization fields. Additionally, for ordinary pedestrians, their daily activity ranges and routes are limited and repetitive, making machine learning particularly suitable for addressing this matching problem, which is undoubtedly a promising direction.

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The final joint filtering primarily includes short-term error matching identification and low-pass filtering, both tailored to the characteristics of error components, effectively filtering out error contributions and enhancing final accuracy and stability.

V. Conclusion

This study proposes a universal pedestrian positioning method suitable for dynamic and varying environments. By organically integrating multiple data sources and algorithms, it achieves high accuracy and stability in continuous seamless positioning. The experimental section thoroughly summarizes and discusses its effectiveness, highlighting the promising prospects of map-matching localization. The localization system in this article can be directly applied and promoted for practical pedestrian positioning, offering significant theoretical and practical value.

Furthermore, the method can be improved in the following directions.

1) *Development of Machine Learning-Based Models:* Investigate machine learning algorithms for map-matching pedestrian localization to enhance the performance of the map-matching component.
2) *Integration of Additional Systems:* Combine more available systems, such as Wi-Fi and Bluetooth, to achieve adaptive subsystem loading and unloading, ultimately forming an adaptive system that supports comprehensive information localization.

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